

Name _____

Date _____

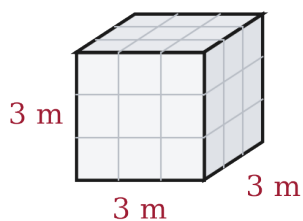
Period _____

Cube Values*Additional Practice 7A.02.03*

1. Complete the table. Each volume is the side length multiplied by itself three times.

Side s	1	2	3	4	5
Volume s^3					
Side $-s$	-1	-2	-3	-4	-5
Volume $(-s)^3$					

2. Find the volume of the cube shown.



3. A cube-shaped box has a volume of 8 cubic centimeters. How long is one edge?

For questions 4 through 7, find the value.

4. $\sqrt[3]{1}$

5. $\sqrt[3]{-27}$

6. $\sqrt[3]{64}$

7. $\sqrt[3]{-125}$

For questions 8 through 10, find every real solution of the equation.

8. $x^3 = 27$

9. $a^3 = -8$

10. $w^3 = 1$

11. Select ALL of the values below that make $x^3 = 125$ true.

☐ 5

☐ -5

☐ $\sqrt[3]{125}$

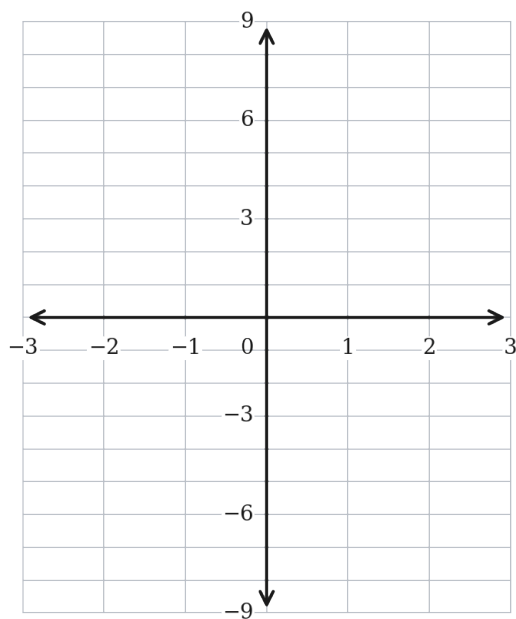
☐ $-\sqrt[3]{125}$

☐ 25

☐ $41\frac{2}{3}$

12. Complete the table of values for $y = x^3$. Then plot the five points on the grid.

x	Substitute into $y = x^3$	Coordinates (x, y)
-2		
-1		
0		
1		
2		



13. The equation $x^2 = 36$ has two solutions, but the equation $x^3 = 27$ has only one. Explain why.

STRETCH. The two questions below use cubes larger than the Grade 8 FAST will ask for. The method is identical — only the arithmetic is bigger.

14. $\sqrt[3]{343}$

15. $m^3 = 512$